

ACHINTYA RAI

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EDUCATION

University of California, San Diego
Bachelor of Science in Computer Engineering

Sept. 2024 – June 2028
GPA: 3.9/4.0

WORK EXPERIENCE

Adobe

Software Engineering Intern

San Jose, CA

June 2026 – Sept. 2026

- Profiled **60.6M+** payment events in Databricks, root-causing **86,387** duplicate records and a **47%** timestamp null rate to establish the authoritative data source for downstream monitoring.
- Diagnosed a **26-hour** processing outage across **1,272** segments as the validation benchmark.
- Designed a time-series anomaly detector benchmarking algorithms across **30+** countries.
- Built SQL and MLOps pipelines automating weekly retraining and hourly inference at scale.

VALT Health

Founding Software Engineer

Remote

Aug. 2025 – Feb. 2026

- Engineered a Flask backend to process **100k+** datapoints from fitness devices (Fitbit, Apple Watch), designing database tables and schemas to unify data formats across **20+** distinct sources.
- Implemented async batch processing in GCP, cutting redundant DB writes **60%**.
- Automated environment setup and data seeding via Bash scripts, cutting testing time up to **80%**.

Decklar (formerly Roambee)

Software Engineering Intern

Sunnyvale, CA

June 2024 – Aug. 2024

- Developed real-time shipment monitoring modules for **Amazon, Coca-Cola, and Samsung**, optimizing Python and NumPy pipelines for a **15%** capacity boost across **1M+** weekly events.
- Implemented advanced data structures for analytics, yielding **20%** faster reporting.
- Refined a Panels-based UI through iterative user testing, improving workflow efficiency by **15%**.

RESEARCH

Graph Theory & Neural Architecture

Undergraduate Research Fellow (NSF REU)

San Diego, CA

March 2026 – Present

- Engineered a Python pipeline analyzing LLM embedding manifolds (Llama-2, DeepSeek-V2), reducing **500GB** checkpoints to **13GB** localized manifolds via weight-matrix extraction utilities.
- Optimized preprocessing for a **30x runtime reduction** on **8GB RAM** Linux systems.
- Applied Rent's Rule and Louvain clustering to raise ICN density to **13.25%**.

Applications of Multi-Agent Systems

Undergraduate Research Fellow (UC Scholars)

San Diego, CA

June 2025 – Aug. 2025

Multi-agent AI Chess Tutor | *Django, React, MCP, SQLite*

- Built an AI chess analysis tool with **Django (Python)**, **React (TypeScript)**, and **SQLite**.
- Architected multi-agent **MCP** workflows, cutting hallucinations **80%**.
- Developed **7 REST API endpoints** and prototypes validated by **15+** chess players and coaches.

PROJECTS

SquaredUpChess | *React, TypeScript, Django, WebSockets, AWS, Docker* 

Aug. 2025 – Dec. 2025

- Engineered a full-stack platform on **AWS** with **20+ Django REST endpoints**.
- Built a **CI/CD** pipeline with automated testing and secure authentication.

MovieRecapinator | *Java, Spring Boot, Maven, JavaScript, IMDb API* 

April 2025

- Built a **Spring Boot** recommendation engine in **Java** processing **700,000+** IMDb entries.

Watchdog | *Python, Asyncio, SQLite, Gemini API* 

April 2026

- Built a concurrent Python CLI with **Asyncio** for real-time URL blacklisting.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, SQL (PostgreSQL/SQLite), Bash, HTML/CSS

Development: Django, Spring Boot, Flask, React, Node.js, FastAPI, JUnit, PyTest

Tools & Cloud: Git, GitHub, Docker, AWS, GCP, Linux/UNIX, GitHub Actions, OAuth